



# DERIVATION HELP



The HELP facility is *not* intended to provide help with the substance of logic. A list of strategic suggestions for doing derivations can be accessed from the Advice button. You can search this document to FIND a particular word or phrase by pressing Ctrl+F.

A brief version of this document, titled “Quick Start”, can be invoked from the Starting button on the module’s button bar.

This HELP facility explains the use of both the elementary features and the advanced tools of the Derivations module. The elementary features are discussed under the headings: [Getting Started](#), [The Derivation Screen](#), [Inserting Symbols and Initiating Operations](#), and [Messages](#). The advanced tools include: the [Available Rules Window](#), [Command Mode](#), [Show Commands](#), [Queuing](#), [Hide Box](#), [Prove to Use – Derived Rules](#), [Theorems as Rules](#), [User Rules](#), [Parameters](#), [Relative Line References](#), [Interchange of Equivalents](#), [Conditional Interchange of Equivalents](#), [Biconditional Derivations](#), [Next-Problem selection](#), [Hover Messages](#), and [Identity & Descriptions](#).

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## GETTING STARTED

### (1) Creating a User by Registering with LOGIC 2010.

When you download the program, it creates an L2KX folder on your computer. This folder holds the program components and also your work files. One of the files has a colored icon and is named "Logic2KX.exe". This file is used to start the program. The L2KX folder also holds a subfolder called "work". This is your Work folder.

The first time you open the program it will ask you for your Institution, and your course. It will then ask you to Register so that it can recognize you as a User of LOGIC 2010. To Register, you must provide your full name, your Student ID# (SID), and your E-Mail address. You will also need to create a Logic Password. Your Name, Institution, and SID appear on all the work you submit. Please enter this information carefully; you will not have the opportunity to revise it (without losing all of your prior work). Be especially careful to enter your correct SID#, it plays a crucial role in identifying you in our database. When you are satisfied with the User Information you have entered, click Save. You will then be asked to create your Logic Password. When you have done so, you will be Registered, and the LOGIC 2010 Menu will open, allowing you to choose a module in which to work. Your newly created Work folder will store all of your work.

### (2) Selecting a problem to work on.

Click the Select button at the bottom of the program window. This will bring up a list of exercises to choose from with the preceding choice highlighted. (A right-click will automatically bring up the next problem in the list.) If you are currently working on a problem, you will be prompted to save your work. Exercises not yet attempted appear in **BLACK**. Exercises done correctly appear in **GREEN**. Exercises done incorrectly (or incompletely) appear in **RED**. To select an exercise to work on, highlight it, and press **Enter**, click **OK**, or simply double-click the exercise. To move through the problem list use the **UpArrow**, **DownArrow**, **PageUp**, and **PageDown** keys. To jump to the top or bottom of the list use the **Home** and **End** keys. If you wish to create your own problem to work on, see [Creating your own 'User' problem](#).

### (3) Creating your own 'User' problem within the Derivations Module.

Rather than working on the preset exercises, you may wish to work on a problem of your own, or on one that your instructor has proposed but has not programmed into the Exercises list. Such problems are called "User" problems. To create a new User problem, simply click the New button on the Derivation screen, and enter the problem. You may either type in the problem using the Keyboard Shortcuts (see [Using the Keyboard](#)) or use the mouse by right-clicking anywhere in the Problem field to invoke the Toolbox (see [Using the Mouse](#)). Anything you type while the

Toolbox is open will be passed through to the line. So you can use a combination of typing and clicking.

If you forget the keyboard shortcut for one of the symbols, allow the mouse to hover over the symbol in the Toolbox, and the keystrokes will appear.

In creating a new problem, remember that each premise, except the last, must be followed by a period, and that the “therefore” symbol, which consists of three dots, must appear between the last premise and the conclusion. The “therefore” symbol appears on a button in the Toolbox, and can also be invoked with **Ctrl+Shift+T**.

User problems are saved to the bottom of the list of exercises in roughly the order in which they were created. Like all other problems, they can be recalled for further work. When you save a User problem, you will be prompted to enter a name. The prefix “User” will be suggested, but any name will work.

#### **(4) Starting a Derivation.**

After selecting an exercise or creating a User problem, press **Enter** or **Alt+Enter** to produce the first line of your derivation. Each line of a derivation contains 5 fields, from left to right: the Line\_Number field (not accessible to users), the Well Formed Formula (WFF) field in which the user enters a WFF, the Justification field in which the user enters a justification for the WFF, the Message field (not accessible to users), and the Explanation field which sometimes produces an Explanation Request button in the form of a question mark. Lines are numbered automatically. If you delete or insert a line, the lines are renumbered automatically (as are the line references in the justifications). For each line, you then provide a formula (in the WFF field) and, to the right, a justification (in the Justification field). Use the Tab key to move back and forth between the WFF field and the Justification field. The program includes a set of editing tools, including a Toolbox of symbols appropriate to each field, to assist you in writing formulas and justifications. The Toolbox is invoked by a right-click of the mouse in the given field (see [Using the Keyboard](#) and [Using the Mouse](#)).

In a typical justification, which invokes a rule and refers to antecedent lines, e.g. “3 7 MP”, the line references must precede the rule name (actually, only the abbreviated rule names are used), and each of the line references and the rule name must be separated from one another by a space or other separator.

For more on derivations see [ORGANIZATION OF THE DERIVATION SCREEN](#), [Command Mode](#), [Queuing](#), and Hide Box.

#### **(5) Checking your work.**

You can check your work at any time by clicking the Check button at the bottom of the Derivation screen, or by using **Ctr l+k** (for “check”). If your derivation is correct, the message “Correct Derivation” will appear in the upper right hand corner as part of the Problem Line.

If your derivation contains errors, a message containing a brief description of the error will appear on each erroneous line. Further explanations can be invoked by clicking the ? button next to the message. Not all messages indicate errors, only messages with a red background indicate errors; others simply provide information.

Even if you correct the errors, the messages will remain until you again check the work. The messages are not saved with the work, so you will also need to recheck your work when a problem that has been saved is recalled (see [Saving your work](#)). For more, see [Messages](#).

#### **(6) Saving your work.**

Save your finished or unfinished work on a problem by clicking the Save button at the bottom of the Derivation screen, or by using **Ctr l+S**. This will save the current version of your work to your Work Disk. If the problem is one that you created, you will be asked to give it a name (see [Creating your own ‘User’ problem](#)). You can confirm that your work has been saved by clicking “Select” and re-selecting the problem. None of the messages are saved with the problem. So when you recall a prior problem, you will have to Check it again.

#### **(7) Submitting your work.**

You can also submit your work over the Internet to a database of homework submissions established for this course. Click the Submit button at the bottom of the screen. You will be asked to verify course information, and to enter a password, which you will create the first time you make a submission. Use **Ctrl+Click** to select (or deselect) more than one exercise to submit. The program will verify that the submitted work has been received.

The database will match your submissions with Assignments constructed by your instructor. You may make multiple submissions of the same problem. The submission that “counts” is the last submission made before the due date of the assignment.

You can view the assignments, and your submissions to each assignment, by clicking “Assignments” on the Menu.

#### **(8) Printing your work.**

**You can print your work at any time by clicking the Print button at the bottom of the Derivation screen, or by using `Ctrl+p`. The printout will contain your user information (name, Student ID#, etc.) as a header. (Note: At many institutions, UCLA among them, if you want to print from a computer in a lab, you must first open a “printing account” at the lab desk.) Use `Ctrl+Click` to select (or deselect) more than one exercise to print. The Derivations Module allows you to print full derivations, one to a page, or to print only results (click the Print Results button). In the latter case, each of the exercises you have selected will be represented by a single line indicating whether the exercise was completed correctly or not.**

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## THE DERIVATION SCREEN

The derivation screen is organized into three parts.

### The Problem Line

At the top is the Problem Line, which contains 3 fields. At the left is the Problem\_Name field. Next comes the Argument field, containing the actual argument to be derived. Very long arguments will wrap. At the right of the Problem Line is the Message field, in which will appear "Correct Derivation", "Incorrect Derivation", etc. when the problem is checked. A few of the preset exercises come with comments ("Try to do this problem in no more than 638 lines"). The comments, when present, will appear above the Problem Line.

### The Button Bar

At the bottom is the Button Bar containing buttons to execute a variety of functions:

| Button                                             | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Select</b><br>(Select Problem)<br><b>Ctrl+o</b> | Used to select a problem to work on. You may select one of the preset exercises or a User problem that you have already created and saved. See <a href="#">Selecting a problem to work on</a> .                                                                                                                                                                                                                                                                                                              |
| <b>User</b><br>(User Problem)                      | <b>Used to create a "User Problem", a new problem of your own that is not in the list of exercises. See <a href="#">Creating your own 'User' problem</a>.</b>                                                                                                                                                                                                                                                                                                                                                |
| <b>Rules</b><br>(Display Available Rules)          | Pops up a window showing all the inference rules. For derived rules, the theorems necessary to derive the rule are indicated. See <a href="#">Available Rules Window</a> .                                                                                                                                                                                                                                                                                                                                   |
| <b>Check</b><br><b>Ctrl+k</b>                      | Initiates an automated check of your work on the current problem, and displays messages describing any errors. See <a href="#">Checking your work</a> .                                                                                                                                                                                                                                                                                                                                                      |
| <b>Save</b><br><b>Ctrl+s</b>                       | Saves your work on the current problem to your Work folder, where it will be available for you to recall it for editing. When you leave the Workspace you will be prompted to save. See <a href="#">Saving your work</a>                                                                                                                                                                                                                                                                                     |
| <b>Delete</b>                                      | Pops up a window asking if you wish to delete the work on the current problem (and begin anew). If the problem is a User problem, you also have the option of deleting the problem itself.                                                                                                                                                                                                                                                                                                                   |
| <b>Submit</b>                                      | Submit one or several problems over the Internet to the course database. After clicking the Submit button you will verify your course, instructor, and TA, enter your logic password, and select the exercises you wish to submit using <b>Ctrl+Click</b> . Then click Submit at the bottom of the selection window to submit all of the selected problems. After a brief delay, you will receive an acknowledgment that the work has been successfully received. See <a href="#">Submitting your work</a> . |

| Button                        | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Print</b><br><b>Ctrl+p</b> | Pops up the Print window, which lists all of the preset exercises along with all of the User problems that you have created and saved. The current problem will be highlighted. Use <b>Ctrl+Click</b> to select (or deselect) more than one exercise to print. The Print button at the bottom will then initiate printing of your work, one problem per page. You may also choose the Print Results button, which causes a single page to be printed showing just a list of the selected problems with the results (Correct or Incorrect) of your work, but without showing the work itself. You can also print a list of exercises (without work) by using the Print List button. See <a href="#">Printing your work</a> . |
| <b>Menu</b>                   | Summon the opening Menu window. The Menu allows you to select a different module or to open another instance of the present module. The Menu module also holds access to the course website, and the Quit button for closing the program. You will be asked if you wish to save your current work before quitting.                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Close</b>                  | Close the present module window. You will be asked if you wish to save your current work before the window is closed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Feedback</b>               | A web-based channel for anonymous or open communication with the Instructor, TAs, and authors of the program. It is especially useful for comments or questions about the program, and is a source for the FAQ facility (see below). If you prefer an individual response over anonymity, include your e-mail address with your message.                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Starting</b>               | A brief and compressed introduction to the Derivations module for beginners.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Help</b>                   | Summons this document.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>FAQ</b>                    | Provides a quick take on Frequently Asked Questions concerning various aspects of the program. Many issues covered in FAQs are not covered in this Help file.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Advice</b>                 | Summon the “Brief Strategic Advice” document. This important document provides strategic advice for doing derivations. See <a href="#">Advice</a> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

## Derivations

### Adding and Deleting Lines

The main part of the screen contains the derivation itself. A derivation is a stack of lines in outline form. New lines are produced by pressing **Enter**, **Alt+Enter**, or some other combination of keys containing **Enter** (see [Enter+<key> combinations](#)).

Lines are deleted by pressing **Alt+Delete**. To insert a line, put the cursor in the line above, and press **Enter** or **Alt+Enter**.

### **Line Numbering**

As noted above, each line of a derivation contains 5 fields: the Line\_Number field, the WFF field, the Justification field, the Message field, and the Explanation field. At the left is the Line\_Number field. Lines are numbered, and renumbered (if a line is inserted or deleted), automatically by the program. When lines are renumbered, the references that appear in the Justification field are automatically recalculated to match. The Line\_Number field is inaccessible to the user.

### **Show Lines**

Next comes the WFF field, in which the formula is displayed. In a Show line, the word "Show" occupies a separate field at the beginning of the WFF field. The Show field is produced by using either the keyboard command, **Ctrl+Shift+S**, or by clicking the Show button in the Toolbox. Both the keyboard command and the Toolbox button are toggles, which insert the Show field if it is not present, and remove it if it is present. It is also possible to simply type the word "Show" into the line. When you press the spacebar, the program will recognize the word, and place it in its own field.

### **Entering Formulas**

A formula can be produced either by using the program's editing tools (see [Using the Keyboard](#) and [Using the Mouse](#)), or, if Command Mode is enabled, by providing a justification that allows the program itself to calculate and insert the formula.

The program accepts both Official and Informal notation.

If the string occupying the WFF field is not a well formed formula (in either the Official or Informal sense), the error message "Parse error at position *n*" (indicating a syntactical problem) or "Lexical error at position *n*" (indicating an unrecognized symbol) is posted immediately. This is an exception to the general principle that error messages are posted only when a Check is called for by the user.

### **Justifications**

The third field is the Justification field in which the content of each line other than a Show line must be justified. The content of this field can be produced by using the keyboard or by using the mouse (as with the WFF field, a Toolbox, containing characters appropriate for justifications, is invoked by a right click anywhere in the Justification field).

In a typical justification, which invokes a rule and refers to antecedent lines, e.g. "3 7 MP", the line references must *precede* the abbreviated rule name, and each of the line references and the rule name must be separated from one another by a separator character. Almost any character other than a letter or a numeral will do as a separator, including, as in the example, a space.

The abbreviations for inference rules can be found in the [Available Rules Window](#), which is invoked by the [Rules](#) button. In addition to the usual abbreviations, the Available Rules window shows abbreviations for sub-rules such as “DNI” for Double Negation Introduction and “DNE” for Double Negation Elimination. The sub-rules are useful for avoiding queries from the program, which would not otherwise know in the case of “ $\sim\sim P$ ” whether you wished to add or remove two negations.

The Available Rules window also lists the Theorems whose instances can be used in derivations. The list of Theorems follows the list of inference rules.

Other abbreviations used in the justification field are:

|                |                                                                                                                                      |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------|
| PR             | Premise (PR1 is the first premise, PR2 the second, and so on)                                                                        |
| ASS CD         | Assumption for Conditional Derivation                                                                                                |
| ASS ID         | Assumption for Indirect Derivation                                                                                                   |
| n DD           | Box and cancel by Direct Derivation, where line n contains the formula in the Show line                                              |
| n CD           | Box and cancel by Conditional Derivation, where line n contains the consequent of the Show line                                      |
| m n ID         | Box and cancel by Indirect Derivation, where lines m and n contain an explicit contradiction                                         |
| n UD           | Box and cancel by Universal Derivation, where line n contains the formula in the Show line without the initial universal quantifiers |
| Tnnn           | Instance of Theorem number nnn (See <a href="#">Prove to Use</a> )                                                                   |
| RTnnn          | The rule corresponding to Theorem number nnn (See <a href="#">Theorems as Rules</a> )                                                |
| Show Conc      | Enter a Show line for the <u>Conclusion</u> of the argument (in the Problem line) (See <a href="#">Show Commands</a> )               |
| Show Cons      | Enter a Show line for the <u>Consequent</u> of the conditional in the immediately preceding uncanceled Show line                     |
| Show Ant n     | Enter a Show line for the <u>Antecedent</u> of the conditional in line n                                                             |
| Show Unneg n   | Enter a Show line for the <u>Unnegation</u> of line n, i.e. the formula whose negation appears in line n                             |
| Show Corr      | Enter a Show line for the <u>Corresponding Conditional</u> to the disjunction in the immediately preceding uncanceled Show line      |
| Show Conj      | Enter a Show line for one of the <u>Conjuncts</u> of the conjunction in the immediately preceding uncanceled Show line               |
| Show Cond      | Enter a Show line for one of the <u>Conditionals</u> of the biconditional in the immediately preceding uncanceled Show line          |
| Show NegDisj n | Enter a Show line for the <u>Negation of one of the Disjuncts</u> of the disjunction in line n                                       |
| Show Inst      | Enter a Show line for the <u>Instance</u> of the universal generalization in the immediately preceding uncanceled Show line          |

Sophisticated users may wish to make use of Justification [Queuing](#), which provides for compressed derivations. In theory, the use of justification queuing alone (*without* the use of the [Hide Box](#) facility) should permit many derivations to be compressed to the point at which only Show lines and box & cancel lines appear, thus, to twice the number of Show lines required.

### **Message Field**

The fourth field in a derivation line is the Message field. It appears to the right of the Justification field. Like the Line\_number field, this field is inaccessible to the user. The messages are provided by the program. (See [MESSAGES](#) for more.)

### **Explanation Button**

Finally, at the far right, there is the fifth field, the Explanation field, in which a  button appears when there is a message in the Message field. Clicking on the  button will open up an Explanation window, which provides additional information about the message that has been posted.

---

## **Inserting Symbols and Initiating Operations**

Most of the symbols that you will need to insert into a derivation can be inserted by using the mouse alone, by using the keyboard alone, or by using the two together. The same is true for many operations, such as checking proofs, boxing and canceling, deleting lines, adding lines, etc.

When entering formulas, outermost parentheses may always be omitted. In general, symbolic sentences in the informal sense are always acceptable to the program.

### **Using the Mouse**

#### **Invoking the Toolbox**

Right click in any area of the line into which you wish to insert the symbols. The Toolbox will appear. Note that the Toolbox that appears in the WFF-field is different from the one that appears for the Justification-field.

Although the toolbox always appears beneath the line from which it was invoked, it can be moved to any portion on your monitor, even outside of the window containing the Logic Program. To move the toolbox, place the mouse pointer in the Title Bar (where the line number appears), and drag and drop. The toolbox cannot be resized.

### **Using the Toolbox**

Each of the symbols you may wish to use should appear on a button in the toolbox. By clicking on each symbol in turn, you can construct the entry for the line. You can also combine clicking with typing. For example, while the WFF toolbox is open, you can type “(P”, then click the “→” button, then type “Q)”. The result will be that “(P→Q)” will appear on the line. The symbols “(”, “P”, “Q”, “)” also appear on buttons in the toolbox, so you can construct the entire WFF just by clicking, if you choose.

Although the cursor is not visible while a toolbox is open, it is there. The symbols you enter will appear at the point the cursor occupies, beginning at the point at which it was when the toolbox was invoked. As usual, the cursor moves to the right each time a character is added. You can also move the invisible cursor using left and right arrow keys as well as **Home** and **End**. The right-click used to invoke the toolbox does not affect the initial position of the cursor.

In addition to symbols the Toolbox contains buttons to add and delete lines, to hide and unhide boxes, and to jump to the other field (Tab).

### **Hover Messages**

All Toolbox buttons other than those that represent normal keyboard symbols contain Hover Messages. These messages appear when the cursor is allowed to linger over a button. The Hover Messages generally indicate the keyboard equivalent of the button. They thus provide a quick and easy way to learn the keyboard shortcuts. Most buttons throughout the program contain Hover Messages.

### **Using the Keyboard**

Most symbols are produced, and operations invoked, by a **Ctrl+Shift+<key>** combination. This means: Hold down the Ctrl and Shift keys while pressing the key associated with the symbol or operation. The key associations are established for easy recollection. They are usually the first letter of the name of the symbol: Ctrl+Shift+C for the conditional sign, Ctrl+Shift+N for the negation sign, etc. The relevant key combination also appears on a hover message when the mouse is used to point at a Toolbox button.

### **Ctrl+Shift+<letter> combinations.**

#### **Ctrl+Shift+<letter> Symbols.**

|                     |   |                 |
|---------------------|---|-----------------|
| <b>Ctrl+Shift+C</b> | → | (Conditional)   |
| <b>Ctrl+Shift+B</b> | ↔ | (Biconditional) |
| <b>Ctrl+Shift+A</b> | ^ | (And)           |
| <b>Ctrl+Shift+O</b> | ∨ | (Or)            |
| <b>Ctrl+Shift+N</b> | ~ | (Negation)      |

|                              |              |                                 |
|------------------------------|--------------|---------------------------------|
| <b>Ctrl+Shift+E</b>          | $\exists$    | (Existential quantifier)        |
| <b>Ctrl+Shift+U</b>          | $\forall$    | (Universal quantifier)          |
| <b>Ctrl+Shift+T</b>          | $\therefore$ | (Therefore)                     |
| <b>Ctrl+Shift+S</b>          | Show         | (toggles "Show" on and off)     |
| <b>= (keyboard 'equals')</b> | =            | (Identity)                      |
| <b>Ctrl+Shift+I</b>          | $\neq$       | (Inequality)                    |
| <b>Ctrl+Shift+D</b>          | $\iota$      | (Definite description operator) |

#### **Ctrl+Shift+<letter> Operations.**

|                       |       |                                   |
|-----------------------|-------|-----------------------------------|
| <b>Ctrl+Shift+X</b>   | boX   | (toggles boX & cancel on and off) |
| <b>Ctrl+[Shift+]K</b> | check | (checkK derivation)               |

Note that while Caps Lock is on, all connectives can still be produced with Ctrl+Shift+<letter> combinations.

Upper case is needed for sentence letters. You may find it convenient to press the Caps Lock key so that all letters are upper case. On a computer keyboard, Caps Lock affects only the letter keys. Shift is still needed to obtain the upper symbol for the keys at the top of the keyboard. When the Caps Lock is on, the Shift key produces lower case letters. To turn Caps Lock off, simply press the key a second time.

#### **Enter+<key> combinations**

Enter+<key> combinations are used primarily for Command Mode and for adding lines.

#### **Enter Operations**

In Command Mode, with the cursor in the Justification field, the justification filled in, and the WFF field empty, Enter will enter the WFF, add a new line, and move to it preparatory to your filling in the next justification. You can complete almost an entire derivation in this way, just filling in justifications and pressing Enter. (Although many Show lines can be entered using Show Commands, some may require 'hand' entry.)

| Combination | Enters formula | Overwrites prior formula | Adds a new line |
|-------------|----------------|--------------------------|-----------------|
| Enter       | YES            | NO                       | YES             |
| Ctrl+Enter  | YES            | YES                      | YES             |
| Shift+Enter | YES            | YES                      | NO              |
| Alt+Enter   | NO             | NO                       | YES             |

Note that Shift+Enter is useful for the final line in a derivation. If Command Mode is not enabled, no formulas are entered.

### **Alt+<key> combinations.**

Alt+<key> combinations are used primarily for deleting, adding, hiding, unhiding, and moving lines.

### **Alt+<key> Operations.**

|                      |                                                                        |
|----------------------|------------------------------------------------------------------------|
| <b>Alt+Delete</b>    | Deletes the line in which the cursor appears                           |
| <b>Alt+Enter</b>     | Adds a blank line below the line in which the cursor appears           |
| <b>Alt+UpArrow</b>   | Hides (rolls up) the subderivation (applicable only in Show lines)     |
| <b>Alt+DownArrow</b> | Unhides (rolls down) the subderivation (applicable only in Show lines) |

To the left of each Show line there is a small box with either a "+" or a "-" in it. By clicking on this box one can Hide or Unhide the subderivation beneath the Show line. The same effect can be obtained using the keyboard by placing the cursor in the Show line and pressing Alt+UpArrow (to Hide) or Alt+DownArrow (to Unhide).

|                        |                                                                |
|------------------------|----------------------------------------------------------------|
| <b>*Alt+RightArrow</b> | Indents line (line becomes subordinate to a lower Show line)   |
| <b>*Alt+RightArrow</b> | Outdents line (line becomes subordinate to a higher Show line) |

**\* Use with great care.** Outdent can affect all lines to which the given line is antecedent; Indent affects the lines above the given line that lie within the same subderivation. It is generally safer to just delete and add new lines. **Alt+Enter** will add a new blank line below the line containing the cursor.

### **Ctrl+<key> combinations.**

### **Ctrl+<key> Editing.**

The usual editing keys are available.

|               |                                                  |
|---------------|--------------------------------------------------|
| <b>Ctrl+c</b> | Copy the highlighted material                    |
| <b>Ctrl+x</b> | Copy and delete the highlighted material         |
| <b>Ctrl+v</b> | Paste the copied material at the cursor location |
| <b>Ctrl+z</b> | Undo the most recent paste or delete             |

The usual alternatives are also available

|                     |                                          |
|---------------------|------------------------------------------|
| <b>Ctrl+Insert</b>  | Copy the highlighted material            |
| <b>Shift+Delete</b> | Copy and delete the highlighted material |

**Shift+Insert**                      Paste the copied material at the cursor location

The program also uses the following throughout all modules.

**Ctrl+k**                                  Check work

**Ctrl+s**                                  Save work

**Ctrl+p**                                  Print work

The *Parse Highlighter* is helpful for formulas that are difficult to parse on the fly.

**Ctrl+e**                                  Highlight wFE

Place the cursor anywhere in a well formed Expression and press **Ctrl+e**. The smallest well formed part containing the cursor will be highlighted. Each subsequent press of **Ctrl+e** will highlight the next smallest part. In this way you can often get a better sense of the parse structure of the expression.

---

## **MESSAGES**

### **Info vs. Error Messages**

There are two kinds of messages: Information (Info) messages, and Error messages. **Info** messages appear against the usual blue background; **Error** messages appear against a red background.

A typical Info message is "**Blank line**". It is not an error to include a blank line, but the program calls it to your attention. Similarly, when a Show line is canceled, the program notes whether it was by a "**Uniform Derivation**" (one in which the assumption matches the basis for boxing and canceling) or by a "**Mixed Derivation**" (one in which they do not). No error is involved in either case.

A typical Error message is "**does not follow from MP**". This message speaks for itself. There are also Error messages to indicate when the string of symbols in the WFF field does not constitute a well formed formula, and when the string of characters in the Justification field does not have a grammatically correct form.

### **The ? Button**

The meaning of each Info and Error message is discussed at greater length in the Explanation screen that is invoked by the **?** button, which appears when there is a message in the Message field. An Explanation screen may also provide an opportunity to allow the program to correct the error. For example, the Explanation screen for "does not follow from MP" explains what the correct result would be, and offers the opportunity for the program to enter the correct result.

### **How Messages are Invoked**

In addition to the two kinds of messages, there are also two ways in which messages are invoked.

The program automatically checks for certain conditions that should be brought to the user's attention immediately. These auto-check conditions include grammatical errors in the WFF field and the Justification field, blank lines, and a few others. Many of these messages report on what may be considered 'typos'. These automatic checks occur when the cursor leaves the relevant field.

Most messages are invoked by the user's request for the derivation to be checked. (Both keyboard and buttons can be used to institute such checking.) Errors of logic, i.e. errors in the use of the system of derivation, are reported following a user-invoked check.

A given line may suffer several infirmities that would generate messages. Because the Message field will only accommodate one message, the messages are arranged in a hierarchy, and only the first of the relevant messages will appear.

---

## DERIVATION TOOLS

### Command Mode

Ordinarily, you enter a formula into the WFF field (the field on the left), and then enter the justification into the Justification field (the field on the right). In Command Mode, you simply enter the justifying annotation into the Justification field, press **Enter**, and the program will insert the appropriate line into the WFF field (and move you down to the Justification field of the next line).

For example, if line 2 is " $(P \rightarrow Q)$ " and line 3 is " $P$ ", then on line 4, you need only type "2 3 MP" into the right hand field and press **Enter**. The program will insert " $Q$ " in the left hand field for you. This can save a lot of typing and editing time. In Command Mode, the only time you need to leave the Justification field is to insert a Show line, and even then, the [Show Commands](#) will often suffice. Should it be necessary to enter a Show line by hand, **Ctrl+Shift+S** will jump you over to the WFF field, insert the word "Show", and set you up to enter the formula to be shown into the Show line. (After entering the formula, **Enter** followed by **Tab** will move you back into position to continue in Command Mode.)

If you give an improper justification, such as attempting to apply MP to " $(P \rightarrow Q)$ " and " $Q$ ", the command cannot be executed. In this case, Command Mode will send an error message and an explanation displaying the fact that the lines you cited do not match the proper form of the premises of MP.

Your Instructor has the ability both to disable Command Mode entirely, and to disable it for specific problems.

## Command Mode Queries

Sometimes, in Command Mode, the program must query you as to your exact intentions. Some rules have more than one form, and more than one form may apply. For example, if you cite a line containing “ $\sim\sim P$ ” and the rule DN, the program cannot know whether you wish two negations to be added or removed. So it presents you with a query.

In some cases, you can avoid a query, by using the sub-rules, whose names are given in the Available Rules window (invoked by the Rules button).

Even aside from querying as to which form of a rule is intended, there are cases like ADD, in which there is simply no way for the program to know what formula you intend to add on. So here again is the occasion for a query in Command Mode.

## Using Sub-rule Names to Avoid Queries

In most cases in which a rule has more than one form, there exist names for each form, though they are less commonly used. For example, DNI is the rule for Double Negation Introduction ( $P \therefore \sim\sim P$ ), and DNE is the rule for Double Negation Elimination ( $\sim\sim P \therefore P$ ). SL is the rule for Simplification on the left, and SR is the rule for Simplification on the right. The names of the forms of each rule can be found in the Available Rules window, which is invoked by clicking the button labeled “Rules” on the main Derivations screen. Use of the sub-rule name will avoid a query regarding the form of the rule.

## Citing Premises in Command Mode

To introduce a premise in Command Mode, when there is more than one premise, you may write just “PR”, in which case the program will query with a menu of premises from which to choose, or, you may add the number of the premise, for example “PR3” for the third premise (there is no separation between “PR” and “3”), to directly enter a particular premise.

---

## Show Commands

Nine special “Show Commands” are provided (four for Chapter I, four more for Chapter II, and one for Chapter III) to allow the Show lines associated with the primary derivation strategies to be entered in Command Mode. Show commands can be used to enter Show lines for the conclusion of the argument, for the consequent of a conditional, for the unnegation of a previous line, for the conditional corresponding to a disjunction, for one of the conjuncts of a conjunction, etc. The strategic use of Show commands is discussed in detail in the Advice document.

A Show Command can be executed only when the immediately preceding uncanceled Show line is of the appropriate form. Thus, “Show Conjunct” is

applicable only if the immediately preceding uncanceled Show line is a conjunction. In this case, a query will appear asking which conjunct you wish to Show.

Show Commands are entered into the Justification field (as are all Command Mode commands) and cannot be combined with (i.e. queued with) any other command.

Show Commands indicate the motivation for entering the Show line. Thus, when a Show line created by the Show command "Show Consequent" is boxed and canceled, a glance at the Show command immediately indicates that the next command should be "-1 CD" (here I use relative references). Similarly, if the just canceled Show line was created by the Show command "Show Unnegation 7", the next command should be "7 R -2 ID" (using a relative reference plus queuing).

To undo a Show Command, put the cursor in the Show line, and press Ctrl+Shift+S, or right-click to bring up the Toolbox, and then click Show/Unshow.

For each Show Command, it suffices to type enough of the initial portion of the command to identify it. Thus, for the command "Show Conjunct", any of the following will suffice: "Show Conj", "Show Conju", "Show Conjun", "Show Conjunc", "Show Conjunct". In the following, brackets [ ] are used to show the optional part of the command.

When beginning a derivation, use

Show Conc[lusion]      To enter a Show line for the Conclusion of the argument.

When the immediately preceding uncanceled Show line is a conditional, and you have already used ASS CD to assume the antecedent, use

Show Cons[equent]      To enter a Show line for the Consequent of the conditional.

When you wish to make use of a conditional, but do not have readily available the antecedent (or the negation of the consequent), use

Show Ant[ecedent] n      To enter a Show line for the Antecedent of the conditional on line n.

When you have already begun an indirect derivation, and wish to make use of the negation of a conditional on line n, use

Show Unneg[ation] n      To enter a Show line for the formula whose negation appears on line n.

When the immediately preceding uncanceled Show line is a disjunction, use

Show Corr[Cond]      To enter a Show line for the Corresponding Conditional. (Alternative strategies are indicated in the Advice document.)

When the immediately preceding uncanceled Show line is a conjunction, use

Show Conj[unct]      To enter a Show line for one of the Conjuncts.

When the immediately preceding uncanceled Show line is a biconditional, use Show Cond[itional] To enter a Show line for one of the Conditionals.

When the immediately preceding uncanceled Show line is a universal generalization, use Show Inst[ance] To enter a Show line for the Instance of the universal generalization required for a UD derivation.

## Queuing

Derivations can be shortened by combining the justifications for a sequence of lines into a 'queued' justification on a single line. For example, a derivation for the argument

$$\sim Q \rightarrow \sim P \quad \therefore P \rightarrow Q$$

might ordinarily look something like this:

|   |                                          |        |
|---|------------------------------------------|--------|
| 1 | <u>Show <math>P \rightarrow Q</math></u> |        |
| 2 | P                                        | ASS CD |
| 3 | $\sim \sim P$                            | 2 DN   |
| 4 | $\sim Q \rightarrow \sim P$              | PR     |
| 5 | $\sim \sim Q$                            | 3 4 MT |
| 6 | Q                                        | 5 DN   |
| 7 | _____                                    | 6 CD   |

However, with queuing, the derivation need occupy only two lines:

|   |                                          |                       |
|---|------------------------------------------|-----------------------|
| 1 | <u>Show <math>P \rightarrow Q</math></u> |                       |
| 2 | _____                                    | ASS CD DN PR MT DN CD |

Both rules and references to lines may be queued. Show lines are the only lines that cannot be combined with other lines in a queue.

To understand how queuing operates, imagine a stack of formulas. Each step in a queued justification either adds a formula to the top of the stack, or applies an inference rule to the appropriate number of formulas drawn from the top of the stack and replaces them with the result, or applies a boxing and canceling rule to the remaining formula (or formulas) on the top of the stack. If the final step is an inference rule, the result is displayed in the WFF field of the line. If the final step is a boxing and canceling rule, the subderivation will be boxed and canceled. In the example above, the first step (ASS CD) places "P" on the stack. The second step applies DN to the formula on the top of the stack, and replaces it with " $\sim \sim P$ ". The third step (PR) adds " $\sim Q \rightarrow \sim P$ " to the top of the stack. The fourth step applies MT to the top two formulas on the stack, and replaces them with " $\sim \sim P$ ". The fifth step applies DN to the formula on the top of the stack, and replaces it with "Q". The sixth step applies the boxing and canceling rule CD to the formula on the top of the stack, and boxes and cancels.

Note that the final step made use of all the remaining formulas on the stack. This is a general requirement, applying to queued and normal justifications: A justification in which more lines are referenced than are used by the rule is erroneous. Thus in line 5 of the original derivation, the justification “2 3 4 MT” would have been erroneous. In a queued justification, it is acceptable to build up extra lines in the stack along the way, so long as they are all used up by the end.

Two other requirements on queued justifications are drawn from those on normal justifications. If a queued justification contains an Assumption (ASS CD or ASS ID), it must be at the beginning of the queue. Just as in a normal justification, an Assumption can only occur as the first step immediately following a Show line. Second, if a queued justification contains a boxing and canceling rule (DD, CD, ID, or UD), it must be at the end of the queue (just as in a normal justification, boxing and canceling can only occur as the final step in a subderivation).

With ingenuity, a derivation can often be reduced to no more than twice the number of Show lines required.

---

### Hide Box

Once a Show line has been canceled, and the associated subderivation boxed, the boxed material is of no further use in the derivation. It continues to take up space, however, and may make it difficult or impossible to survey all the relevant elements of a derivation by bringing them all on screen at the same time. To assist with this problem, the program provides a facility to hide lines that are already boxed.

Once a Show line has been canceled, the entire boxed subderivation can be hidden by clicking on the little minus sign (“-”) immediately to the left of the word **Show**. Upon doing so, the minus sign changes to a plus sign (“+”), indicating that hidden lines are present. Note also that the line numbering indicates that some lines are missing from the screen. You can click the plus sign to unhide the hidden lines.

Keyboard shortcuts to hide and unhide are **Alt+UpArrow** to Hide, **Alt+DownArrow** to Unhide. Hide and Unhide buttons also appear on the Toolbox. The keyboard shortcuts and the Toolbox buttons operate only on Show lines.

---

### Prove to Use – Derived Rules

Unless specifically enabled by the Instructor, derived rules cannot be used until the underlying theorem has been proved. For example, the derived rule CDJ (Conditional Disjunction), which has the form:  $\sim P \rightarrow Q \therefore P \vee Q$ , is based on Theorem T45:  $(\sim P \rightarrow Q) \leftrightarrow (P \vee Q)$ . To use CDJ you must first prove T45. Once T45

is proved, one could, at worst, copy the proof of the relevant instance into any new proof, apply BC to obtain  $(\sim P \rightarrow Q) \rightarrow (P \vee Q)$ . Then, if one had already obtained  $(\sim P \rightarrow Q)$ , MP would yield the desired result. Hence the use of derived rules, only shortens proofs; it does not allow anything to be proved that could not be proved using only the primitive rules. (A few unmentioned details must be established to give a rigorous argument for this fact.) Both the Problem List, accessed through the Select button, and the Available Rules window, accessed through the Rules button, indicate which theorems must be proved to activate each of the derived rules.

In most cases, it will be obvious how the theorem justifies the rule. However, in a few cases, we have allowed a theorem to justify a rule that really assumes both the theorem and a simple variant. This is usually when the proof of the variant is obvious from the proof of the theorem. A simple example is the use of T70 to justify *both* SSimp1 and SSimp2, when properly speaking it immediately justifies only SSimp1. Part of our reason for doing this is that we did not wish to disrupt the numbering of the theorems in the text.

The same constraint of a prior proof applies to the use of any theorem in a derivation. The TerryText states the rule allowing the use of theorems as follows: "At any step, an instance of a previously proved theorem may be entered with the name of the theorem given as justification. (e.g. 'T32')" (Chapter 2, p 32). The program enables exactly this rule. You must first prove a theorem in order to use it in subsequent derivations.

In proving theorems in order to use them or to use derived rules based on them, it is *not* required that the theorems be proved in any particular order. At any stage, any theorems that are proved are usable, as are any derived rules based on those theorems.

If, in doing a derivation, it is desirable to use a rule that has not yet been proved, it is possible to summon the menu (by using the Menu button), open a second instance of the Derivation module, prove the relevant theorem in the new instance of the derivation module, then save it and return to the original module to complete the original derivation with the use of the rule. A proof must be saved for the relevant rule or theorem to be usable.

## **Theorems as Rules**

Any conditional or biconditional theorem can be used as a rule (once the theorem is proved) by prefixing "R" to the theorem name. For example, once theorem T20:  $(P \rightarrow \sim P) \rightarrow \sim P$ , is proved, the corresponding rule RT20:  $P \rightarrow \sim P \therefore \sim P$ , is automatically enabled.

It is often considerably more convenient to use the corresponding rule rather than a theorem. Use of a theorem requires the user to provide the scheme of substitution manually, but when a rule is used, the program can usually figure out the substitution on its own.

Biconditional theorems yield two forms of the corresponding rule. Thus theorem T115:  $(P \rightarrow \sim P) \leftrightarrow \sim P$ , yields two forms for the rule RT115:  $P \rightarrow \sim P \therefore \sim P$ , and  $\sim P \therefore P \rightarrow \sim P$ . Since the two sides of most biconditional theorems have different forms, usually only one form of the rule will be applicable (and the program can figure this out). However, for those cases in which the user wishes to specify the form used, the suffixes “L” and “R” indicate, respectively, the left to right form, and the right to left form. Thus “RT115L” refers only to:  $P \rightarrow \sim P \therefore \sim P$ , and “RT115R” refers only to:  $\sim P \therefore P \rightarrow \sim P$ .

If the rule corresponding to a theorem has a conjunction as a premise, an alternative form of the rule containing each of the conjuncts as separate premises is also enabled. Thus, theorem 88:  $(P \wedge \sim Q) \rightarrow \sim(P \leftrightarrow Q)$  yields two forms for the rule RT88:  $(P \wedge \sim Q) \therefore \sim(P \leftrightarrow Q)$ , and  $P. \sim Q \therefore \sim(P \leftrightarrow Q)$ . The latter form avoids having to first apply ADJ.

## User Rules

You may find that certain patterns of inference that take several steps to justify are frequently repeated, and you may therefore wish to have a single rule as a shortcut. Some such rules already exist, and can be viewed in the [Available Rules Window](#). These rules are enabled by proving the indicated theorem. However, you may not find the inference pattern you wish to use. In this case, you can create your own User Rule. The steps are simple:

1. Open the User Problem screen by clicking the User button.
2. Use the Toolbox (right click) or keyboard to fill in the problem in a way to mirror the inference pattern of the rule you wish to use, for example, “ $\sim P \rightarrow \sim Q. Q \therefore P$ ”. Don’t include the quotes, and remember to add a period, “.”, after each premise but the last, and a Therefore-sign, “ $\therefore$ ” between the premises and the conclusion. (On the keyboard, Ctrl+Shift+T inserts the Therefore-sign.)
3. Give a derivation for the problem, and check it to make sure it is correct.
4. Save the problem with a name beginning with the two letters “UR” (case doesn’t matter), for example, “urMTDN”. You can use any letter or numeral or character on the top row of keys (! @ # \$ % ^ & \* ( ) \_ + - =), but no spaces.
5. The new User Rule should now appear at the top of the [Available Rules Window](#), and can be used by citing the name you gave it, including the mandatory “UR” prefix.

6. User Problems are stored at the bottom of the Select list. To delete a User Rule, select and load the User Problem with that name, then delete it by clicking the Delete button (at the bottom of the screen). When queried, choose, "Delete this problem?".
  7. There is at present no way to rename a User Rule other than by creating a new rule and deleting the old, so choose your User Rule names carefully.
- 

## Using Parameters

Certain Command Mode commands invariably invoke queries. For example, UI will always invoke a query asking what the instantial term should be, and similarly for EI. ADD will always invoke a query asking for the formula to be added. The use of parameters allows one to answer these questions directly in the justification field without waiting for the query to appear.

A parameter is introduced by a forward slash, /. There must be no spaces or other separators between the rule name and the slash or between the slash and the actual parameter.

Thus if one wishes to UI line 7 to the variable "z", one would write:

7 UI/z

Any of the following will be accepted:

7 EI/b

7 ADDR/~(P $\wedge$ Q)

The parameter will fill the first text field to appear in any query. Thus if one forgets the exact rule for addition on the right (ADDR), one could simply write:

7 ADD/~(P $\wedge$ Q)

This will require a query as to whether to add on the right or the left, but when that query is answered, there will be no subsequent query for the formula.

If a rule were to require more than one parameter, for example if we had a rule of the form

DBLADD P  $\therefore$  Q $\vee$ P $\vee$ R

The two parameters would follow one another, each introduced by a slash

7 DBLADD/Q/R

When there is more than one parameter, the order in which the parameters should appear in the justification field is the order in which they would be asked for in a query. This is usually the left to right order in which they appear in the resulting WFF.

---

## Relative Line References

It is often the case that line references refer to the immediately preceding (legitimate) line or lines. These lines can be referred to by their position relative to the line whose justification is being given. Thus the immediately preceding line is line -1, the line before that is -2, and so forth. Because only legitimate antecedent lines (those not boxed and not containing uncanceled show) are counted, when a boxing and canceling has taken place, the newly canceled Show line will become line -1. Thus in the standard derivation of a biconditional by deriving each of the two conditionals, the justification for the CB line will usually read: -1 -2 CB.

Relative line references are used only to speed the entry of references, and to reduce errors. When the command is executed the relative references are replaced by traditional, absolute references.

---

## Rules Window

This window, invoked by clicking the Rules button, lists all rules, including [User Rules](#), along with the abbreviations used to cite them. The Justification field accepts only abbreviations. “3 7 MP” is recognized; “3 7 Modus Ponens” is not. The Available Rules window is merely a list. Rule names cannot be inserted into justifications from this window.

The Rules window can now remain open while you work, and when you move from exercise to exercise. The cost of this is that the window no longer tracks exactly which rules and Theorems are enabled for the current exercise. Instructors have the ability to enable and disable specific rules for all problems or for individual problems. Unless specifically enabled or disabled by the Instructor, all primitive rules are enabled, and all derived rules are enabled only after the relevant theorems have been proved. (See [Prove to Use](#).) The Rules window also indicates, for each derived rule, which theorems must be proved to enable it.

---

## Interchange of Equivalents

The rule of Interchange of Equivalents (IE) is one of the most powerful rules in logic, and is the only rule that allows for the direct logical manipulation of arbitrary subformulas. It is commonly assumed in sophisticated discussions.

Interchange of Equivalents is usually formulated in a way to depend on a *theorem* as the basis for the interchange (hence Interchange of *Equivalents*). In our formulation, the interchange can be also based on any so-called “material equivalents”, that is, any antecedent line that is a universally closed biconditional. Universal closure is only required for variables that are bound by external

quantifiers or that would be picked up by external quantifiers after the interchange is made.

Thus to replace " $R(xyz)$ " by " $\sim S(xzu)$ " in " $\forall y \exists u (Fu \wedge R(xyz))$ ", the antecedent biconditional " $R(xyz) \leftrightarrow \sim S(xzu)$ " would have to be universally closed for " $y$ " and for " $u$ ", but it need not be closed for " $x$ " or for " $z$ ". Hence, an antecedent line, " $\forall y \forall u (R(xyz) \leftrightarrow \sim S(xzu))$ " would justify the transition from " $\forall y \exists u (Fu \wedge R(xyz))$ " to " $\forall y \exists u (Fu \wedge \sim S(xzu))$ ". The justification cites the line into which the interchange is being made, the rule "IE", and, if desired, the biconditional parameter "[the biconditional rule, premise, or antecedent line number]". The program will pop up a dialog box displaying the line into which the interchange is to be made, and asking you to highlight the subformula to be replaced. It may then offer some alternatives depending on which side of the biconditional is to be replaced. (For example, if you propose to replace the " $\sim \sim Fu$ " in " $\forall y \exists u (\sim \sim Fu \wedge R(xyz))$ " using the biconditional DN theorem, the program would have to query whether to add or remove the two negations.)

As a practical matter, the biconditional parameter will usually be a biconditional theorem (the name of the corresponding rule can be used as a parameter). In this case universal closure for arbitrary variables is assumed. Thus, to use a commutativity law to replace " $(Fu \wedge R(xyz))$ " by " $(R(xyz) \wedge Fu)$ " in " $\forall y \exists u (Fu \wedge R(xyz))$ " (occurring in line 7), the justification would read "7 IE/COM".

This rule is usually enabled or disabled by Instructors rather than it being given on a Prove to Use basis. A large number of theorems would be required to enable IE by the Prove to Use technique, and even then the justification of IE requires a more complex argument than that for justifying the rule corresponding to a single conditional theorem. In sentential logic, T96, T101-104, and certain analogs to T102-104 would be required. Your instructor may wish to sketch the justification of IE for the language of Chapter II based on those theorems.

---

### Conditional Interchange of Equivalents

The rule of Conditional Interchange of Equivalents (CIE) is a strengthened form of Interchange of Equivalents. It allows the interchange to be made on the basis of a *conditional* biconditional (for example, an instance of " $\sim Q \rightarrow (P \vee Q \leftrightarrow P)$ ") when the antecedent of the conditional is already available as a theorem or antecedent line. The antecedent of the conditional becomes a second parameter. As in IE, universal closure is required for variables that are bound by external quantifiers or that would be picked up by external quantifiers after the interchange is made.

Here again, as a practical matter, both the conditional biconditional and its antecedent will often be theorems (as with IE, the names of the corresponding rules can be used as parameters). Thus, to use the above cited conditional biconditional,

which happens to correspond to the rule SSimp3 to replace “ $(R(xyz) \vee (Fu \wedge \sim Fu))$ ” by “ $R(xyz)$ ” in “ $\forall y \exists u (R(xyz) \vee (Fu \wedge \sim Fu))$ ” (occurring in line 7), the justification would read “7 CIE/SSimp/T36” (T36 is “ $\sim (P \wedge \sim P)$ ”). The program will query as to which subformula is to be replaced, and as to the instance of T36 you intend. (It may then offer some alternatives depending on which side of the biconditional is to be replaced, as in the DN case described in the preceding section.)

The primary use of the 16 derived rules of Sentential Simplification (SSimp) is as a basis for CIE. They are especially useful in simplifying the constructions in ?DNF and ?NOV problems (regarding which, see below). In these applications, instances of T36 and T59 are often used as the premise/antecedent of the SSimp rule. The use of CIE is somewhat more complex than for simple IE, but it is very satisfying when needed.

---

### Biconditional Derivation

This derived rule can only be enabled by an Instructor. It is a technique for deriving biconditionals. “ASS BD” [or “ASS BDR” or “ASS BDL”] allows the user to assume one side of the biconditional to be shown. The following steps must all be justified by the rule IE, which is *reversible* in that the result of applying the rule also yields the line to which the rule was applied. When the other side of the biconditional (the side not assumed) is obtained, one may cite the line and box and cancel by “BD”.

No forms of “mixed” biconditional derivations are allowed.

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### Advice

Strategic advice is a document suggesting basic strategies for constructing derivations. This is not yet a context sensitive facility.

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### Identity and Descriptions

The program covers Chapters V (Identity) and VI (Descriptions) of the text. The ‘equals’ key in the top row of the keyboard is used for the identity symbol, and **Ctrl+Shift+I** is used to produce the *inequality* sign, “ $\neq$ ”. **Ctrl+Shift+D** is used to produce the definite descriptions operator, “ $\iota$ ”, which appears in the text as an inverted iota.

As usual, the proper abbreviations for all rules can be seen by clicking the Rules button. The rule of Self-Identity is “SI $\iota$ D”. The rule for Proper Descriptions is “PrD”. The rule for Improper Descriptions is “ImD”.

---

### Wild Cards: “?DNF”, “?CNF”, “?PNX” and “?NOV”

The expressions “?PNX” and “?NOV” function like formulas, except they can only occur in Show lines, and when the Show is Boxed and Canceled, “?PNX” will match any formula in prenex form, and “?NOV” will match any formula without overlay of quantifiers. They are used to create exercises in which the User must construct the matching formula rather than having the formula given as part of the problem.

Additional Wild Cards are: “?DNF” and “?CNF”. “?DNF” will match any formula in Disjunctive Normal Form. A formula is in DNF if it contains no conditionals or biconditionals, no negations, conjunctions, or disjunctions in the *immediate* scope of a negation, and no disjunctions in the *immediate* scope of a conjunction. (This permits quantifiers provided the formulas within the scopes of the quantifiers are in DNF.) “?CNF” will match any formula in Conjunctive Normal Form. A formula is in CNF if it contains no conditionals or biconditionals, no negations, conjunctions, or disjunctions in the *immediate* scope of a negation, and no conjunctions in the *immediate* scope of a disjunction. (This permits quantifiers provided the formulas within the scopes of the quantifiers are in CNF.)

In doing these problems, the formulas can become very long and difficult to parse on the fly. It is often helpful to invoke the *Parse Highlighter* by using **Ctr l+e**. If you place the cursor anywhere in a well formed expression and press **Ctr l+e**, the smallest well formed part containing the cursor will be highlighted. Each subsequent press of **Ctr l+e** will highlight the next smallest part. In this way you can often get a better sense of the pieces you need to work on.

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